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Homework 3

Solutions

1) Given models,

$$\text{Total effect } y_i = \beta_{11} + \beta_{12}x_i + \epsilon_i$$

$$\text{Mediator model } m_i = \beta_{21} + \beta_{22}x_i + \epsilon_i$$

$$\text{Direct + Indirect effect } y_i = \beta_{31} + \beta_{32}m_i + \beta_{33}x_i + \epsilon_i$$

$$\text{Variance of } \epsilon_i \Rightarrow \sigma^2 = 1$$

$$\text{Significance level } \alpha = 0.05$$

Null hypothesis of Sobel test:

The Sobel test checks whether the indirect (mediation) effect is zero.

$$\Rightarrow H_0: \beta_{22}\beta_{32} = 0$$

That is, no mediation effect through m

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1) 2) Using the given data, we need to regress the models and find the estimators for (β_{11}, β_{12}) , (β_{21}, β_{22}) & $(\beta_{31}, \beta_{32}, \beta_{33})$

Using OLS with intercept,

Model 1: y on x ,

$$y_i = \beta_{11} + \beta_{12} x_i + \epsilon_i$$

Code:

```
model1 <- lm(y ~ x, data = data)
```

```
summary(model1)
```

$$\Rightarrow y_i = -1.607 + 2.638 x_i + \epsilon_i$$

$$\Rightarrow (\beta_{11}, \beta_{12}) = (-1.607, 2.638)$$

Model 2: m on x

$$m_i = \beta_{21} + \beta_{22} x_i + \epsilon_i$$

Code:

```
model2 <- lm(m ~ x, data = data)
```

```
summary(model2)
```




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$$\Rightarrow m_i = -0.772 + 2.055x_i + \epsilon_i$$

$$\Rightarrow (\beta_{21}, \beta_{22}) = (-0.772, 2.055)$$

Model 3: y on m and x

$$y_i = \beta_{31} + \beta_{32}m_i + \beta_{33}x_i + \epsilon_i$$

Code: $\Gamma \text{ model 3} \leftarrow \text{lm}(y \sim m + x, \text{data} = \text{data})$

Summary (model 3)

$$\Rightarrow y_i = -0.666 + 1.219m_i + 0.133x_i + \epsilon_i$$

$$\Rightarrow (\beta_{31}, \beta_{32}, \beta_{33}) = (-0.666, 1.219, 0.133)$$

1) 3) To find

Variance of $\hat{\beta}_{22}$ & $\hat{\beta}_{32}$

We know that

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$



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In R, we can retrieve this using `Vcov()`.

Code: `Vcov(model 2)`

```
var_b22 <- Vcov(model 2)[“x”, “x”]
```

```
var_b22 # Extract var of slope(x)
```

~~$\Rightarrow \text{Var}(\hat{\beta}_{22}) = 0.00138$~~

$$\Rightarrow \text{Var}(\hat{\beta}_{22}) = 0.00138 //$$

Similarly, for $\hat{\beta}_{32}$,

Code: ~~`Vcov`~~ `Vcov(model 3)`

```
var_b32 <- Vcov(model 3)[“m”, “m”]
```

```
var_b32 # Extract var of slope  
or coeff for (m)
```

$$\Rightarrow \text{Var}(\hat{\beta}_{32}) = 0.00869 //$$

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1)4) To find

Sobel z-test statistic,

$$Z = \frac{\hat{\beta}_{22} \hat{\beta}_{32}}{\sqrt{\hat{\beta}_{32}^2 \text{Var}(\hat{\beta}_{22}) + \hat{\beta}_{22}^2 \text{Var}(\hat{\beta}_{32})}}$$

Substituting the values we derived in previous sections, we get,

$$Z = \frac{2.055 * 1.219}{\sqrt{(1.219)^2 * 0.00138 + (2.055)^2 * 0.00869}}$$

$$\Rightarrow Z = 12.7303 //$$

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1) 5) To find
p-value & conclusion of the test

We can find the 2-sided p-value as follows.

R code:

```
p-value ← 2 * (1 - pnorm(abs(z)))  
p-value
```

⇒ p-value ≈ 0 .

As $p < 0.05$, $\rightarrow$ Reject H_0 .

⇒ There is a statistically significant
mediation effect

Conclusion:

⇒ 'm' mediates the effect of 'x' on 'y'.

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2) 1) Given

$$\text{Model: } y_i = x_i \beta + \epsilon_i$$

ϵ is unknown

Data is centered (no intercept)

$$\sum_{i=1}^n x_i^2 = h$$

The least square estimator minimizes

$$RSS(\beta) = \sum_{i=1}^n (y_i - x_i \beta)^2$$

$$= \sum_{i=1}^n y_i^2 - 2\beta \sum_{i=1}^n x_i y_i + \beta^2 \sum_{i=1}^n x_i^2$$

Take derivative with respect to β ,

$$\frac{d}{d\beta} RSS(\beta) = -2 \sum_{i=1}^n x_i y_i + 2\beta \sum_{i=1}^n x_i^2 = 0$$

$$\Rightarrow \beta \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$



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$$\Rightarrow \hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

We know that $\sum_{i=1}^n x_i^2 = n$

$$\Rightarrow \hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i y_i$$

Hence proved

To find it using R,

Rcode: $\beta_hat \leftarrow \text{mean}(\text{data}\$x * \text{data}\$y)$
 β_hat

$$\Rightarrow \hat{\beta} = 79.144 \quad (\text{for our data})$$

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2) 2) As we know,

$$RSS = \sum_{i=1}^n (y_i - x_i \hat{\beta})^2$$

$$= \sum_{i=1}^n y_i^2 - 2\hat{\beta} \sum_{i=1}^n x_i y_i + \hat{\beta}^2 \sum_{i=1}^n x_i^2$$

From the previous derivation,

$$\sum_{i=1}^n x_i y_i = \hat{\beta} \sum_{i=1}^n x_i^2$$

$$\Rightarrow RSS = \sum_{i=1}^n y_i^2 - 2\hat{\beta} \left(\hat{\beta} \sum_{i=1}^n x_i^2 \right) + \hat{\beta}^2 \sum_{i=1}^n x_i^2$$

$$= \sum_{i=1}^n y_i^2 - \hat{\beta}^2 \sum_{i=1}^n x_i^2$$

We know that $\sum_{i=1}^n x_i^2 = n$

$$\Rightarrow RSS = \sum_{i=1}^n y_i^2 - n \hat{\beta}^2$$

Hence proved



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For our data,

R code: $n \leftarrow \text{nrow}(\text{data})$

$$\text{rss} \leftarrow \text{sum}(\text{data} \$ y^2) - n * \text{beta_hat}^2$$

rss

$$\Rightarrow \text{rss} = -607041.938 //$$

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2)3) To find,

F-statistic for testing,

$$H_0: \beta = 0$$

We know that,

$$F_{\text{stat}} = \frac{\text{MSR}}{\text{MSE}}$$

$$\text{MSR} = \frac{\text{SSR}}{\text{Reg of freedom}}$$

Reg of freedom

(Here regression degrees of freedom = 1)

$$\Rightarrow \text{MSR} = \text{SSR}$$



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As data is centered & no intercept,

Total Sum of Squares,

$$TSS = \sum_{i=1}^n y_i^2$$

We know,

Residual Sum of Squares,

$$RSS = \sum_{i=1}^n y_i^2 - n\hat{\beta}^2$$

Regression Sum of Square,

$$\Rightarrow SSR = TSS - RSS$$

$$= \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i^2 - n\hat{\beta}^2 \right)$$

$$= n\hat{\beta}^2$$

$$MSE = \frac{RSS}{\text{Residual degrees of freedom}}$$

$$= \frac{RSS}{n-1}$$

$$\Rightarrow F_{\text{stat}} = \frac{MSR}{MSE} = \frac{n\hat{\beta}^2}{RSS/(n-1)}$$



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The estimator of σ^2 ,

$$\hat{\sigma}^2 = \frac{RSS}{n-1}$$

$$\Rightarrow F_{stat} = \frac{n \hat{\beta}^2}{\hat{\sigma}^2}$$

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2) 4) If H_0 is true, Find the distribution of $\hat{\beta}$

We know that,

OLS estimator,

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i y_i$$

Substituting the model $y_i = x_i \beta + \sigma \epsilon_i$,

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i (x_i \beta + \sigma \epsilon_i)$$

$$\Rightarrow \hat{\beta} = \frac{\beta}{n} \sum_{i=1}^n x_i^2 + \frac{\sigma}{n} \sum_{i=1}^n x_i \epsilon_i$$



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Using $\sum x_i^2 = h$,

$$\hat{\beta} = \beta + \frac{\sigma}{n} \sum_{i=1}^n x_i \varepsilon_i$$

Since $E[\varepsilon_i] = 0$, $E(\hat{\beta}) = \beta$

$$\text{Var}(\hat{\beta}) = \text{Var}\left(\frac{\sigma}{n} \sum_{i=1}^n x_i \varepsilon_i\right)$$

$$= \frac{\sigma^2}{n^2} \sum_{i=1}^n x_i^2 \text{Var}(\varepsilon_i)$$

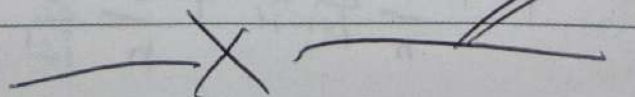
Since $\text{Var}(\varepsilon_i) = 1$,

$$\text{Var}(\hat{\beta}) = \frac{\sigma^2}{n^2} \cdot h = \frac{\sigma^2}{n}$$

If $H_0: \beta = 0$

$$\Rightarrow \hat{\beta} \sim N\left(0, \frac{\sigma^2}{n}\right)$$

If null hypothesis is true, then
distribution of $\hat{\beta}$ is $N\left(0, \frac{\sigma^2}{n}\right)$





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2)5) Given $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

From prob 2),

$$RSS = \sum_{i=1}^n (y_i - x_i \hat{\beta})^2$$

$$\Rightarrow \hat{\sigma}^2 = \frac{RSS}{n-1} \quad \text{as } \hat{y}_i = x_i \hat{\beta}$$

Using distribution from previous section,

$$\frac{\hat{\beta} - \beta}{\sqrt{\hat{\sigma}^2/n}} \sim N(0, 1)$$

Replace σ^2 with $\hat{\sigma}^2$, we get.

$$T = \frac{\sqrt{n} \hat{\beta}}{\hat{\sigma}}$$

Under H_0 : $T \sim t_{n-1}$



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We also derived earlier,

$$F = \frac{n \hat{\beta}^2}{\hat{\sigma}^2}$$

$$\Rightarrow F = T^2$$

Since $T \sim t_{n-1}$

$$\Rightarrow T^2 \sim F_{1, n-1}$$

$\Rightarrow$ t-test & F-test are equivalent,

$\Rightarrow$ They produce identical p-values & decisions.

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